

Exponential dispersion models

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Last time: Exponential dispersion models

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Example $Y \sim N(\mu, \sigma^2)$

Exponential dispersion models

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Example $Y \sim \text{Bernoulli}(p)$

EDM components

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Cumulants and the cumulant generating function

Activity

Work on the activity (handout).